

9(a)

any **eight** from:

discontinuous

- 1 one / few, genes control a phenotype ;
- 2 qualitative ;
- 3 discrete categories / no intermediates ;
- 4 different alleles at single gene locus have large effect on phenotype ;
- 5 different genes have different effects ;
- 6 little / no, contribution by environment to phenotype
- 7 example ; e.g. albinism / sickle cell anaemia / haemophilia / Huntington's disease

continuous

- 8 several genes control a phenotype ;
- 9 quantitative ;
- 10 range of categories / many intermediates ;
- 11 different alleles at single gene locus have small effects ;
- 12 environment has considerable influence on phenotype ;
- 13 example ; e.g. height / mass

8

Question	Answer	Marks
9(b)	any seven from: 1. choose parents with good features ; 2. breed these ; 3. repeat for many generations ; 4. introduction of disease resistance ; 5 named crop disease ; 6 dwarf varieties ; 7 (dwarf varieties) mutant alleles for gibberellin synthesis ; 8 (dwarf varieties) more energy put into grain than into height (of plant) ; 9 (dwarf varieties) less susceptible to being knocked over by weather ; 10 inbreeding leads to uniformity ; 11 named example ; e.g. standard height / cobs ready to harvest at same time 12 hybridisation leads to hybrid vigour ;	7